

Topics: 14.6 Directional Derivatives and the Gradient Vector; 14.7 Maximum and Minimum Values
Please put away all electronic devices, including cell phones and calculators.

This content is protected and may not be shared, uploaded, or distributed.

1. Find the directional derivative of the function at the given point in the indicated direction.

(a) $f(x, y) = \frac{x}{y^2}$ at the point $P(3, -1)$, in the direction from P toward $Q(-2, 11)$.

The direction vector is

$$\overrightarrow{PQ} = \langle -2 - 3, 11 - (-1) \rangle = \langle -5, 12 \rangle.$$

Since

$$|\overrightarrow{PQ}| = \sqrt{(-5)^2 + 12^2} = 13,$$

the corresponding unit vector is

$$\mathbf{u} = \left\langle -\frac{5}{13}, \frac{12}{13} \right\rangle.$$

Next, we compute the gradient:

$$\nabla f(x, y) = \left\langle \frac{1}{y^2}, -\frac{2x}{y^3} \right\rangle.$$

Evaluating at $(3, -1)$, we get

$$\nabla f(3, -1) = \langle 1, 6 \rangle.$$

Therefore, the directional derivative is

$$D_{\mathbf{u}}f(3, -1) = \nabla f(3, -1) \cdot \mathbf{u} = \langle 1, 6 \rangle \cdot \left\langle -\frac{5}{13}, \frac{12}{13} \right\rangle = \frac{67}{13}.$$

- (b) $g(x, y) = x \ln y$ at the point $(2, e)$, in the direction that makes an angle of $\frac{\pi}{6}$ above the positive x -axis.

A unit vector in the direction $\frac{\pi}{6}$ above the positive x -axis is

$$\mathbf{u} = \left\langle \cos \frac{\pi}{6}, \sin \frac{\pi}{6} \right\rangle = \left\langle \frac{\sqrt{3}}{2}, \frac{1}{2} \right\rangle.$$

Next, we compute the gradient:

$$\nabla g(x, y) = \left\langle \ln y, \frac{x}{y} \right\rangle.$$

Evaluating at $(2, e)$, we get

$$\nabla g(2, e) = \left\langle 1, \frac{2}{e} \right\rangle.$$

Therefore, the directional derivative is

$$D_{\mathbf{u}}g(2, e) = \nabla g(2, e) \cdot \mathbf{u} = \left\langle 1, \frac{2}{e} \right\rangle \cdot \left\langle \frac{\sqrt{3}}{2}, \frac{1}{2} \right\rangle = \frac{\sqrt{3}}{2} + \frac{1}{e}.$$

- (c) $h(x, y, z) = x^2y^3 - 4xz$ at the point $(1, 1, 1)$ in the direction of the vector $\mathbf{v} = \langle -1, 2, 0 \rangle$.

First, we find the unit vector in the direction of $\mathbf{v}$. Since

$$|\mathbf{v}| = \sqrt{(-1)^2 + 2^2 + 0^2} = \sqrt{5},$$

we have

$$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} = \left\langle -\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, 0 \right\rangle.$$

Next, we compute the gradient:

$$\nabla h(x, y, z) = \langle 2xy^3 - 4z, 3x^2y^2, -4x \rangle.$$

Evaluating at $(1, 1, 1)$, we get

$$\nabla h(1, 1, 1) = \langle -2, 3, -4 \rangle.$$

Therefore, the directional derivative is

$$D_{\mathbf{u}}h(1, 1, 1) = \nabla h(1, 1, 1) \cdot \mathbf{u} = \langle -2, 3, -4 \rangle \cdot \left\langle -\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, 0 \right\rangle = \frac{8}{\sqrt{5}}.$$

2. Find the maximum rate of change of the function at the given point and the direction in which it occurs.

(a) $f(x, y) = \sin(xy)$ at the point $(1, 0)$.

We first compute the gradient:

$$\nabla f(x, y) = \langle f_x, f_y \rangle = \langle y \cos(xy), x \cos(xy) \rangle.$$

Evaluating at $(1, 0)$, we get

$$\nabla f(1, 0) = \langle 0 \cos(0), 1 \cos(0) \rangle = \langle 0, 1 \rangle.$$

The maximum rate of change is the magnitude of the gradient:

$$|\nabla f(1, 0)| = \sqrt{0^2 + 1^2} = 1.$$

The direction in which it occurs is the direction of the gradient, namely $\langle 0, 1 \rangle$.

(b) $g(x, y, z) = e^{2x} \cos(y - 2z)$ at the point $(1, \pi, 0)$.

We first compute the gradient:

$$\nabla g(x, y, z) = \langle g_x, g_y, g_z \rangle = \langle 2e^{2x} \cos(y - 2z), -e^{2x} \sin(y - 2z), 2e^{2x} \sin(y - 2z) \rangle.$$

Evaluating at $(1, \pi, 0)$, we get

$$\nabla g(1, \pi, 0) = \langle 2e^2 \cos(\pi), -e^2 \sin(\pi), 2e^2 \sin(\pi) \rangle = \langle -2e^2, 0, 0 \rangle.$$

The maximum rate of change is the magnitude of the gradient:

$$|\nabla g(1, \pi, 0)| = \sqrt{(-2e^2)^2} = 2e^2.$$

The direction in which it occurs is the direction of the gradient, so the unit direction vector is

$$\langle -1, 0, 0 \rangle.$$

3. Find an equation of the tangent plane and parametric equations of the normal line to the surface $xy^2z^3 = 8$ at the point $P(2, 2, 1)$.

Let $F(x, y, z) = xy^2z^3$. Then the surface $xy^2z^3 = 8$ is the level surface $F(x, y, z) = 8$, so the gradient ∇F is normal to the surface.

We first compute the gradient:

$$\nabla F(x, y, z) = \langle y^2z^3, 2xyz^3, 3xy^2z^2 \rangle.$$

Evaluating at $P(2, 2, 1)$, we get

$$\nabla F(2, 2, 1) = \langle 4, 8, 24 \rangle.$$

Therefore, an equation of the tangent plane at $P(2, 2, 1)$ is

$$4(x - 2) + 8(y - 2) + 24(z - 1) = 0.$$

A normal line through $P(2, 2, 1)$ has direction vector $\nabla F(2, 2, 1) = \langle 4, 8, 24 \rangle$, so it has parametric equations

$$x = 2 + 4t, \quad y = 2 + 8t, \quad z = 1 + 24t.$$

4. For each function, find all critical points. Then use the Second Derivatives Test to determine whether the function has a local maximum, a local minimum, or a saddle point there.

(a) $f(x, y) = 4x^2 + 9y^2 + 8x - 36y + 24$

We first compute the first order partial derivatives:

$$f_x(x, y) = 8x + 8, \quad f_y(x, y) = 18y - 36.$$

To find the critical points, we set both partial derivatives equal to zero:

$$8x + 8 = 0, \quad 18y - 36 = 0.$$

Solving gives that the only critical point is $(-1, 2)$.

Next we compute the second order partial derivatives:

$$f_{xx}(x, y) = 8, \quad f_{yy}(x, y) = 18, \quad f_{xy}(x, y) = 0.$$

Recall that

$$D(x, y) = f_{xx}(x, y)f_{yy}(x, y) - (f_{xy}(x, y))^2.$$

At the critical point,

$$D(-1, 2) = 8 \cdot 18 - 0^2 = 144 > 0.$$

Since also

$$f_{xx}(-1, 2) = 8 > 0,$$

f has a local minimum at $(-1, 2)$.

(b) $g(x, y) = (y - 2)x^2 - y^2$

We first compute the first order partial derivatives:

$$g_x(x, y) = 2x(y - 2), \quad g_y(x, y) = x^2 - 2y.$$

To find the critical points, we set both partial derivatives equal to zero:

$$2x(y - 2) = 0, \quad x^2 - 2y = 0.$$

From $2x(y - 2) = 0$, we have either $x = 0$ or $y = 2$.

If $x = 0$, then the equation $x^2 - 2y = 0$ gives $y = 0$, so one critical point is $(0, 0)$.

If $y = 2$, then the equation $x^2 - 2y = 0$ gives

$$x^2 - 4 = 0,$$

so $x = \pm 2$. This gives the critical points $(2, 2)$ and $(-2, 2)$.

Therefore, the critical points are

$$(0, 0), \quad (2, 2), \quad (-2, 2).$$

Next we compute the second order partial derivatives:

$$g_{xx}(x, y) = 2(y - 2), \quad g_{yy}(x, y) = -2, \quad g_{xy}(x, y) = 2x.$$

At $(0, 0)$, we have

$$D(0, 0) = (-4)(-2) - 0^2 = 8 > 0.$$

Since also

$$g_{xx}(0, 0) = -4 < 0,$$

g has a local maximum at $(0, 0)$.

At $(2, 2)$, we have

$$D(2, 2) = (0)(-2) - 4^2 = -16 < 0,$$

so $(2, 2)$ is a saddle point.

At $(-2, 2)$, we have

$$D(-2, 2) = (0)(-2) - (-4)^2 = -16 < 0,$$

so $(-2, 2)$ is a saddle point.

(c) $h(x, y) = (x + y^2)e^x$

We first compute the first order partial derivatives:

$$h_x(x, y) = e^x(x + y^2 + 1), \quad h_y(x, y) = 2ye^x.$$

To find the critical points, we set both partial derivatives equal to zero:

$$e^x(x + y^2 + 1) = 0, \quad 2ye^x = 0.$$

Since $e^x \neq 0$, these equations simplify to

$$x + y^2 + 1 = 0, \quad y = 0.$$

Substituting $y = 0$ into the first equation gives

$$x + 1 = 0,$$

so $x = -1$. Thus the only critical point is $(-1, 0)$.

Next we compute the second order partial derivatives:

$$h_{xx}(x, y) = e^x(x + y^2 + 2), \quad h_{yy}(x, y) = 2e^x, \quad h_{xy}(x, y) = 2ye^x.$$

At the critical point,

$$D(-1, 0) = \left(\frac{1}{e}\right) \left(\frac{2}{e}\right) - 0^2 = \frac{2}{e^2} > 0.$$

Since also

$$h_{xx}(-1, 0) = \frac{1}{e} > 0,$$

h has a local minimum at $(-1, 0)$.

5. Find the absolute maximum and absolute minimum values of the function f on the set D .

(a) $f(x, y) = x^2 + y^2 + x^2y + 9$, where $D = \{(x, y) \in \mathbb{R}^2 : |x| \leq 1 \text{ and } |y| \leq 1\}$

Since f is continuous on the closed and bounded set D , its absolute extrema must occur either at a critical point in the interior of D or on the boundary of D .

We compute

$$f_x(x, y) = 2x + 2xy = 2x(1 + y), \quad f_y(x, y) = 2y + x^2.$$

Setting these equal to zero, we get

$$2x(1 + y) = 0, \quad 2y + x^2 = 0.$$

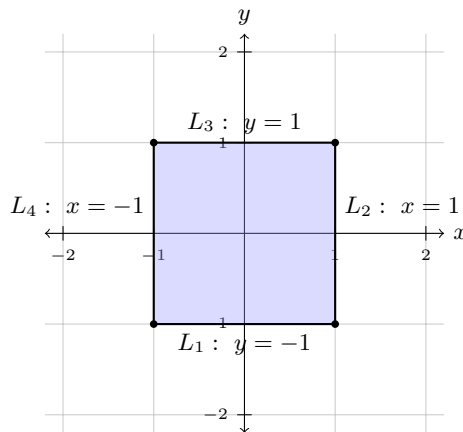
If $x = 0$, then $2y + 0 = 0$, so $y = 0$. If $y = -1$, then

$$2(-1) + x^2 = 0 \Rightarrow x^2 = 2,$$

which is not possible in D since $|x| \leq 1$. Thus the only critical point in D is $(0, 0)$, and here we have

$$f(0, 0) = 0^2 + 0^2 + 0^2(0) + 9 = 9.$$

Next we check the boundary.



- On L_1 : $y = -1$, where $-1 \leq x \leq 1$, so

$$f(x, -1) = x^2 + (-1)^2 + x^2(-1) + 9 = 10.$$

Thus on L_1 , both the minimum and maximum are

$$f(x, -1) = 10.$$

- On L_2 : $x = 1$, where $-1 \leq y \leq 1$, so

$$f(1, y) = 1 + y^2 + y + 9 = y^2 + y + 10.$$

We compute

$$\frac{d}{dy}(y^2 + y + 10) = 2y + 1.$$

Setting this equal to zero gives

$$2y + 1 = 0 \Rightarrow y = -\frac{1}{2}.$$

Thus we check $y = -1$, $y = 1$, and $y = -\frac{1}{2}$:

$$f(1, -1) = 10, \quad f(1, 1) = 12, \quad f\left(1, -\frac{1}{2}\right) = \frac{39}{4}.$$

So on L_2 , the minimum is

$$f\left(1, -\frac{1}{2}\right) = \frac{39}{4},$$

and the maximum is

$$f(1, 1) = 12.$$

- On L_3 : $y = 1$, where $-1 \leq x \leq 1$, so

$$f(x, 1) = x^2 + 1 + x^2(1) + 9 = 2x^2 + 10.$$

We compute

$$\frac{d}{dx}(2x^2 + 10) = 4x.$$

Setting this equal to zero gives

$$4x = 0 \Rightarrow x = 0.$$

Thus we check $x = -1$, $x = 1$, and $x = 0$:

$$f(-1, 1) = 12, \quad f(1, 1) = 12, \quad f(0, 1) = 10.$$

So on L_3 , the minimum is

$$f(0, 1) = 10,$$

and the maximum is

$$f(\pm 1, 1) = 12.$$

- On L_4 : $x = -1$, where $-1 \leq y \leq 1$, so

$$f(-1, y) = 1 + y^2 + y + 9 = y^2 + y + 10.$$

We compute

$$\frac{d}{dy}(y^2 + y + 10) = 2y + 1.$$

Setting this equal to zero gives

$$2y + 1 = 0 \Rightarrow y = -\frac{1}{2}.$$

Thus we check $y = -1$, $y = 1$, and $y = -\frac{1}{2}$:

$$f(-1, -1) = 10, \quad f(-1, 1) = 12, \quad f\left(-1, -\frac{1}{2}\right) = \frac{39}{4}.$$

So on L_4 , the minimum is

$$f\left(-1, -\frac{1}{2}\right) = \frac{39}{4},$$

and the maximum is

$$f(-1, 1) = 12.$$

Comparing all values, the absolute minimum is

$$f(0,0) = 9,$$

and the absolute maximum is

$$f(\pm 1, 1) = 12.$$

- (b) $f(x, y) = x^2 + y^2 - 2x$, where D is the closed triangular region with vertices $(2, 0)$, $(0, 2)$, and $(0, -2)$.

Since f is continuous on the closed and bounded set D , its absolute extrema must occur either at a critical point in the interior of D or on the boundary of D .

We compute

$$f_x(x, y) = 2x - 2, \quad f_y(x, y) = 2y.$$

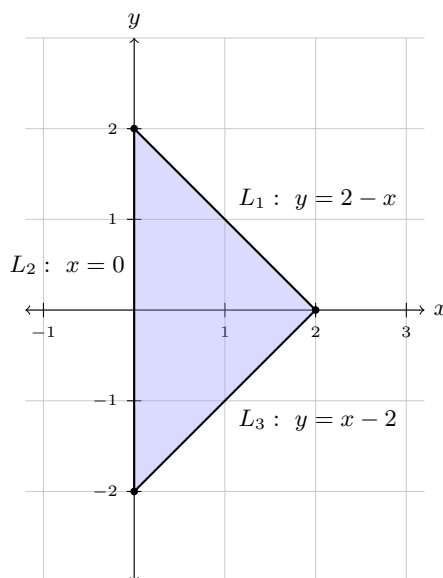
Setting these equal to zero, we get

$$2x - 2 = 0, \quad 2y = 0.$$

Thus $x = 1$ and $y = 0$, so the only critical point in D is $(1, 0)$, and here we have

$$f(1, 0) = 1^2 + 0^2 - 2(1) = -1.$$

Next we check the boundary.



- On $L_1: y = 2 - x$, where $0 \leq x \leq 2$, so

$$f(x, 2 - x) = x^2 + (2 - x)^2 - 2x = 2x^2 - 6x + 4.$$

We compute

$$\frac{d}{dx}(2x^2 - 6x + 4) = 4x - 6.$$

Setting this equal to zero gives

$$4x - 6 = 0 \quad \Rightarrow \quad x = \frac{3}{2}.$$

Thus we check $x = 0$, $x = 2$, and $x = \frac{3}{2}$:

$$f(0, 2) = 4, \quad f(2, 0) = 0, \quad f\left(\frac{3}{2}, \frac{1}{2}\right) = -\frac{1}{2}.$$

So on L_1 , the minimum is

$$f\left(\frac{3}{2}, \frac{1}{2}\right) = -\frac{1}{2},$$

and the maximum is

$$f(0, 2) = 4.$$

- On L_2 : $x = 0$, where $-2 \leq y \leq 2$, so

$$f(0, y) = y^2.$$

This has minimum at $y = 0$, with

$$f(0, 0) = 0,$$

and maximum at the endpoints $y = \pm 2$, with

$$f(0, \pm 2) = 4.$$

- On L_3 : $y = x - 2$, where $0 \leq x \leq 2$, so

$$f(x, x - 2) = x^2 + (x - 2)^2 - 2x = 2x^2 - 6x + 4.$$

We compute

$$\frac{d}{dx}(2x^2 - 6x + 4) = 4x - 6.$$

Setting this equal to zero gives

$$4x - 6 = 0 \Rightarrow x = \frac{3}{2}.$$

Thus we check $x = 0$, $x = 2$, and $x = \frac{3}{2}$:

$$f(0, -2) = 4, \quad f(2, 0) = 0, \quad f\left(\frac{3}{2}, -\frac{1}{2}\right) = -\frac{1}{2}.$$

So on L_3 , the minimum is

$$f\left(\frac{3}{2}, -\frac{1}{2}\right) = -\frac{1}{2},$$

and the maximum is

$$f(0, -2) = 4.$$

Comparing all values, the absolute minimum is

$$f(1, 0) = -1,$$

and the absolute maximum is

$$f(0, \pm 2) = 4.$$

6. A tennis ball manufacturer's monthly profit (in thousands of dollars) is modeled by

$$f(x, y) = 48x + 96y - x^2 - 2xy - 9y^2,$$

where x is the number of tennis balls manufactured (in thousands) and y is the number of hours of advertising purchased. The company can produce at most 50,000 tennis balls per month, and it can purchase at most 25 hours of advertising. Find the values of x and y that maximize profit, and find the maximum profit.

Let

$$D = \{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq 50, 0 \leq y \leq 25\}.$$

Since f is continuous on the closed and bounded set D , the absolute maximum of f must occur either at a critical point in the interior of D or on the boundary of D .

We compute

$$f_x(x, y) = 48 - 2x - 2y, \quad f_y(x, y) = 96 - 2x - 18y.$$

Setting these equal to zero, we get

$$48 - 2x - 2y = 0, \quad 96 - 2x - 18y = 0.$$

Equivalently,

$$x + y = 24, \quad x + 9y = 48.$$

Subtracting gives

$$8y = 24 \Rightarrow y = 3,$$

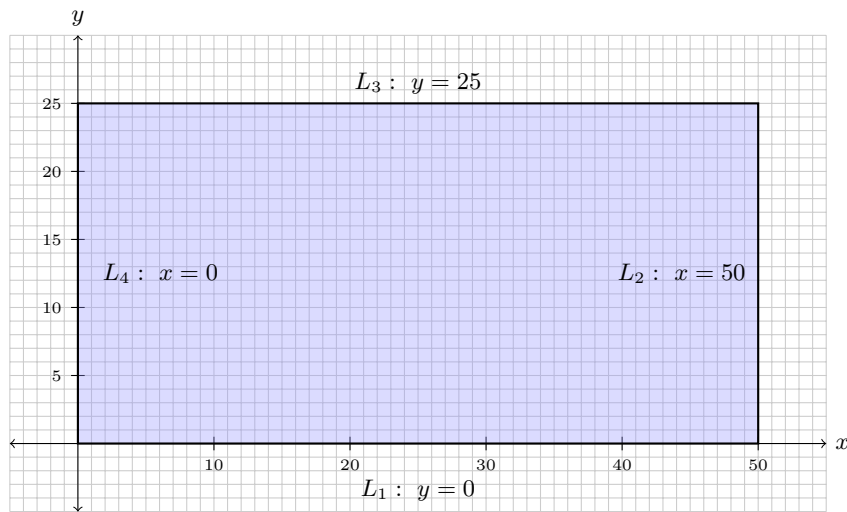
and then

$$x = 24 - 3 = 21.$$

Thus the only critical point in D is $(21, 3)$, and here we have

$$f(21, 3) = 48(21) + 96(3) - 21^2 - 2(21)(3) - 9(3)^2 = 648.$$

Next we check the boundary.



- On $L_1: y = 0$, where $0 \leq x \leq 50$, so

$$f(x, 0) = 48x - x^2.$$

We compute

$$\frac{d}{dx}(48x - x^2) = 48 - 2x.$$

Setting this equal to zero gives

$$48 - 2x = 0 \Rightarrow x = 24.$$

Thus we check $x = 0$, $x = 50$, and $x = 24$:

$$f(0, 0) = 0, \quad f(50, 0) = -100, \quad f(24, 0) = 576.$$

So on L_1 , the maximum is

$$f(24, 0) = 576.$$

- On L_2 : $x = 50$, where $0 \leq y \leq 25$, so

$$f(50, y) = 48(50) + 96y - 50^2 - 2(50)y - 9y^2 = -100 - 4y - 9y^2.$$

We compute

$$\frac{d}{dy}(-100 - 4y - 9y^2) = -4 - 18y.$$

Setting this equal to zero gives

$$-4 - 18y = 0 \Rightarrow y = -\frac{2}{9},$$

which is not in the interval $0 \leq y \leq 25$. Thus we check only the endpoints:

$$f(50, 0) = -100, \quad f(50, 25) = -5825.$$

So on L_2 , the maximum is

$$f(50, 0) = -100.$$

- On L_3 : $y = 25$, where $0 \leq x \leq 50$, so

$$f(x, 25) = 48x + 96(25) - x^2 - 2x(25) - 9(25)^2 = -x^2 - 2x - 3225.$$

We compute

$$\frac{d}{dx}(-x^2 - 2x - 3225) = -2x - 2.$$

Setting this equal to zero gives

$$-2x - 2 = 0 \Rightarrow x = -1,$$

which is not in the interval $0 \leq x \leq 50$. Thus we check only the endpoints:

$$f(0, 25) = -3225, \quad f(50, 25) = -5825.$$

So on L_3 , the maximum is

$$f(0, 25) = -3225.$$

- On L_4 : $x = 0$, where $0 \leq y \leq 25$, so

$$f(0, y) = 96y - 9y^2.$$

We compute

$$\frac{d}{dy}(96y - 9y^2) = 96 - 18y.$$

Setting this equal to zero gives

$$96 - 18y = 0 \Rightarrow y = \frac{16}{3}.$$

Thus we check $y = 0$, $y = 25$, and $y = \frac{16}{3}$:

$$f(0, 0) = 0, \quad f(0, 25) = -3225, \quad f\left(0, \frac{16}{3}\right) = 256.$$

So on L_4 , the maximum is

$$f\left(0, \frac{16}{3}\right) = 256.$$

Comparing all values, the absolute maximum is

$$f(21, 3) = 648.$$

Therefore, the profit is maximized when the company manufactures 21 thousand tennis balls and purchases 3 hours of advertising. The maximum profit is \$648,000.

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Directional Derivatives. Let f be differentiable at a point P , and let $\mathbf{u}$ be a unit vector. The directional derivative of f at P in the direction of $\mathbf{u}$ is given by

$$D_{\mathbf{u}}f(P) = \nabla f(P) \cdot \mathbf{u}.$$

The directional derivative is the rate of change of f at P in the direction of $\mathbf{u}$. If $\nabla f(P) \neq \mathbf{0}$, then the maximum value of the directional derivative at P is $|\nabla f(P)|$ and this maximum occurs when $\mathbf{u}$ has the same direction as $\nabla f(P)$. In other words, $\nabla f(P)$ points in the direction of the fastest increase of f at P .

Tangent Planes to Level Surfaces. Suppose a surface S is expressed as a level surface

$$F(x, y, z) = k,$$

where F is differentiable, and let (x_0, y_0, z_0) be a point on S . Then the gradient vector $\nabla F(x_0, y_0, z_0)$ is orthogonal to the level surface at (x_0, y_0, z_0) , provided that $\nabla F(x_0, y_0, z_0) \neq \mathbf{0}$. Therefore, the tangent plane to the surface at (x_0, y_0, z_0) is the plane through that point with normal vector $\nabla F(x_0, y_0, z_0)$, and its equation is

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0.$$

Critical Points and Local Extrema. Let f be a function of two variables, and let (a, b) be a point in the domain of f . We say that f has a *local maximum* at (a, b) if $f(a, b) \geq f(x, y)$ for all points (x, y) near (a, b) . A *local minimum* is defined analogously. We say that (a, b) is a *critical point* of f if $f_x(a, b) = 0$ and $f_y(a, b) = 0$, or if at least one of these partial derivatives does not exist at (a, b) . If f has a local maximum or local minimum at (a, b) , then (a, b) is necessarily a critical point. A point (a, b) at which $f_x(a, b) = f_y(a, b) = 0$ and f has neither a local maximum nor a local minimum is called a *saddle point*.

The Second Derivatives Test. Suppose that $f_x(a, b) = f_y(a, b) = 0$ and that the second partial derivatives of f are continuous near (a, b) . Define

$$D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - (f_{xy}(a, b))^2.$$

Then:

- If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$, then f has a local minimum at (a, b) .
- If $D(a, b) > 0$ and $f_{xx}(a, b) < 0$, then f has a local maximum at (a, b) .
- If $D(a, b) < 0$, then f has a saddle point at (a, b) .
- If $D(a, b) = 0$, the test is inconclusive.

Absolute Extrema on Closed and Bounded Sets. If f is continuous on a closed and bounded set D in $\mathbb{R}^2$, then f attains an absolute maximum value and an absolute minimum value on D . To find these absolute extrema, evaluate f at all critical points in the interior of D , and at all candidates for extrema on the boundary of D . The largest of these values is the absolute maximum value of f on D , and the smallest is the absolute minimum value.

Suggested Textbook Problems

Section 14.6: 1-44, 47-56

Section 14.7: 1-56